

Mobile Application Development

Lecture 02

Today's Agenda

- Introduction to Android Environment
- Setting up Environment
- Application Fundamentals
 - Android Platform Architecture

Development Environment

- Java is the most popular language used to develop Android applications.
- Android devices don't run .class and .jar files directly.
- Instead, to improve speed and battery performance
 - Android devices use their own optimized formats for compiled code.
- That means that you can't use an ordinary Java development environment
- You also need special tools to convert your compiled code into an Android format, to deploy them to an Android device, and to let you debug the app once it's running

The Android SDK

The Android Software Development Kit contains the libraries and tools you need to develop Android apps. Here are some of the main points:

SDK Platform

There's one of these for each version of Android.

SDK Tools

Tools for debugging and testing, plus other useful utilities. The SDK also features a set of platform dependent tools.

Sample apps

If you want practical code examples to help you understand how to use some of the APIs, the sample apps might help you.



Documentation

So you can access the latest API documentation offline.

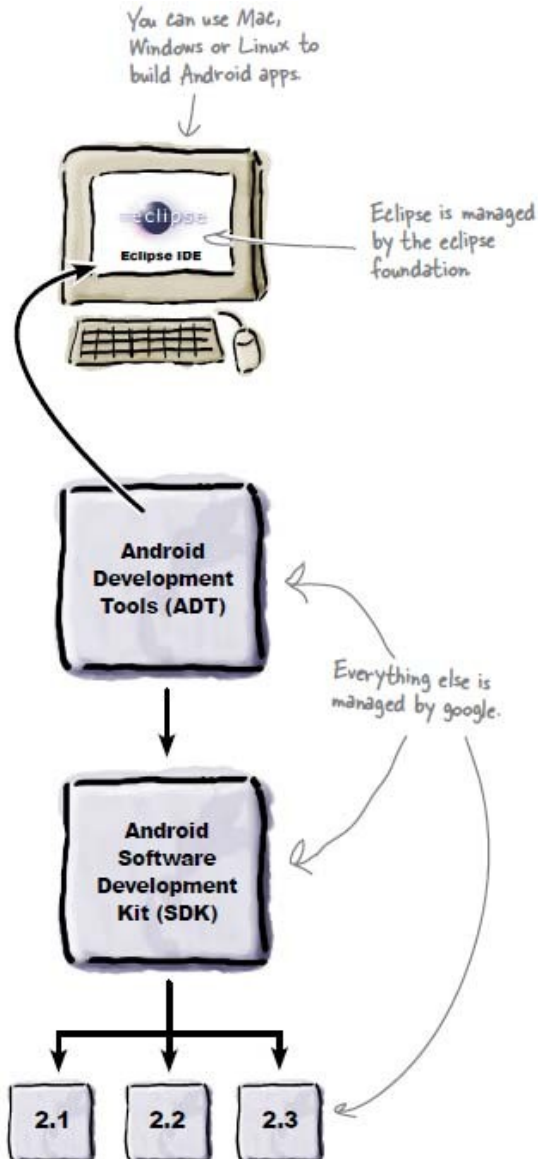
Android support

Extra APIs that aren't available in the standard platform.

Google Play Billing

Allows you to integrate billing services in your app.

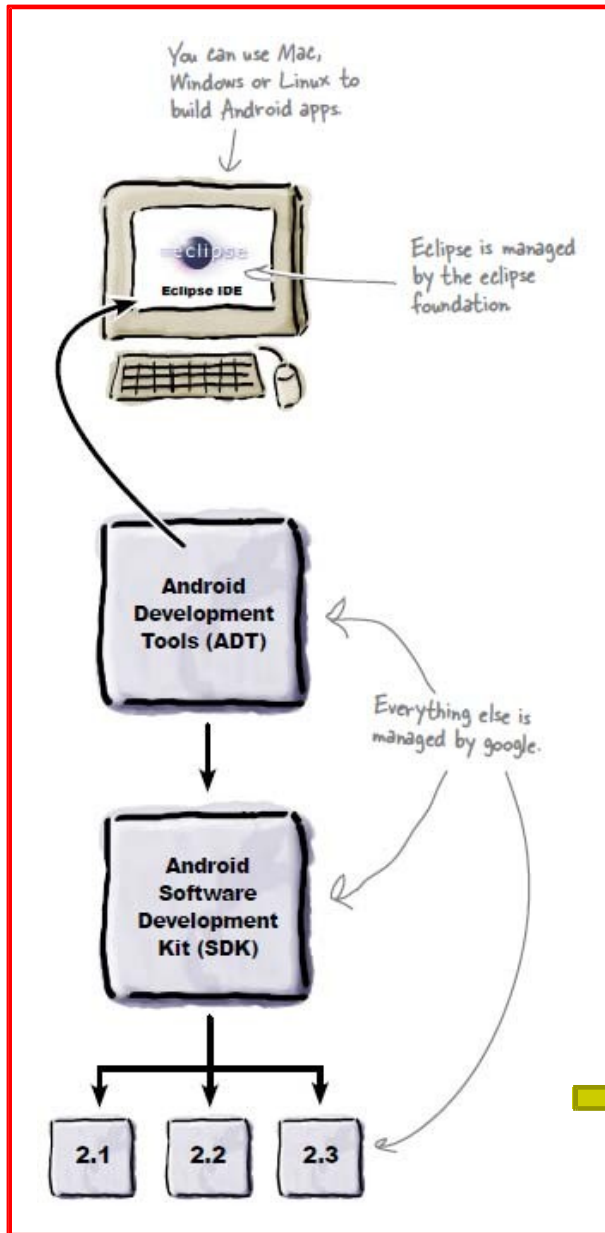
Old Developer Android Environment



- **Eclipse IDE:** type code in, compile, not Android-specific
- **Android Dev Tools (ADT):** Eclipse plugin, adds Android functionality
- **Android Software Dev Kit (SDK):** Tools to build, test and run apps
- **Packages:** Enables developing for various Android versions

New Developer Android Environment

- Google developed it's own IDE called
- **Android Studio**
- Combines tools in old development environment into 1 IDE
- Cleaner interface specifically for Android Development (e.g. drag and drop app design)
- In December 2014, Google announced it will stop supporting Eclipse IDE



Installing Android Studio

The most important place to find information and links to tools is the Android Developers site:

<http://developer.android.com>

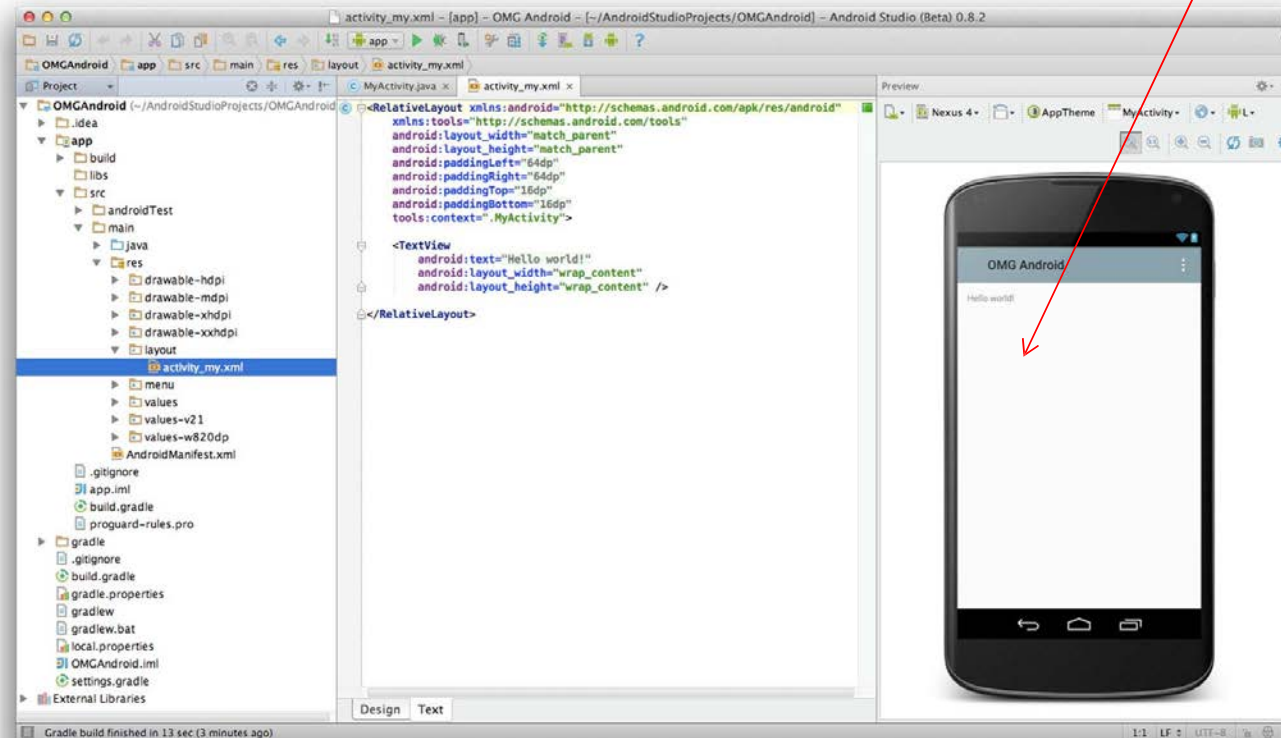
Installing Android Studio

- **Step 1:** Install Java (at least version 1.7)
 - **Note:** You may already have Java installed. Check first
 - Can find the JDK at <http://www.oracle.com/technetwork/java/javase/downloads/index.html>
- **Step 2:** Set JAVA_HOME system environmental variable
 - This variable tells applications that need Java where it is installed
- **Step 3:** Install Android Studio (version 4 is latest)
<https://developer.android.com/studio/install>
- Bucky Roberts (thenewboston): nice youtube Android tutorials
 - **Tutorial 1:** Install Java <https://www.youtube.com/watch?v=QAbQgLGKd3Y>
 - **Tutorial 2:** Install Android Studio <https://www.youtube.com/watch?v=zEsDwzjPJ5c>

Where to Run Android App

- Android app can run on:
 - Real phone (or device)
 - Emulator (software version of phone)

Preview phone
pane in Android
Studio



Emulator Pros and Cons

- Pros:
 - Conveniently test app on basic hardware by clicking in software
 - Easy to test app on various devices (phones, tablets, TVs, etc), various screen sizes
- Cons:
 - Some hardware missing, especially hardware for sensing environment
 - E.g. GPS, camera, video recording, etc
 - Needs a lot of memory to run and eventually slows down the system

Emulator Limitations

- No support for
 - Phone calls (calling or receiving)
 - USB connections
 - Camera/video capture (input)
 - Bluetooth
 - Sensors, accelerometer, gyroscope, etc
 - Device-attached headphones
 - Determining battery charge level and AC charging state
 - Determining SD card insert/eject
- Slow!!!

Running Android App on RealPhone

- Need USB cord to copy app from development PC to phone



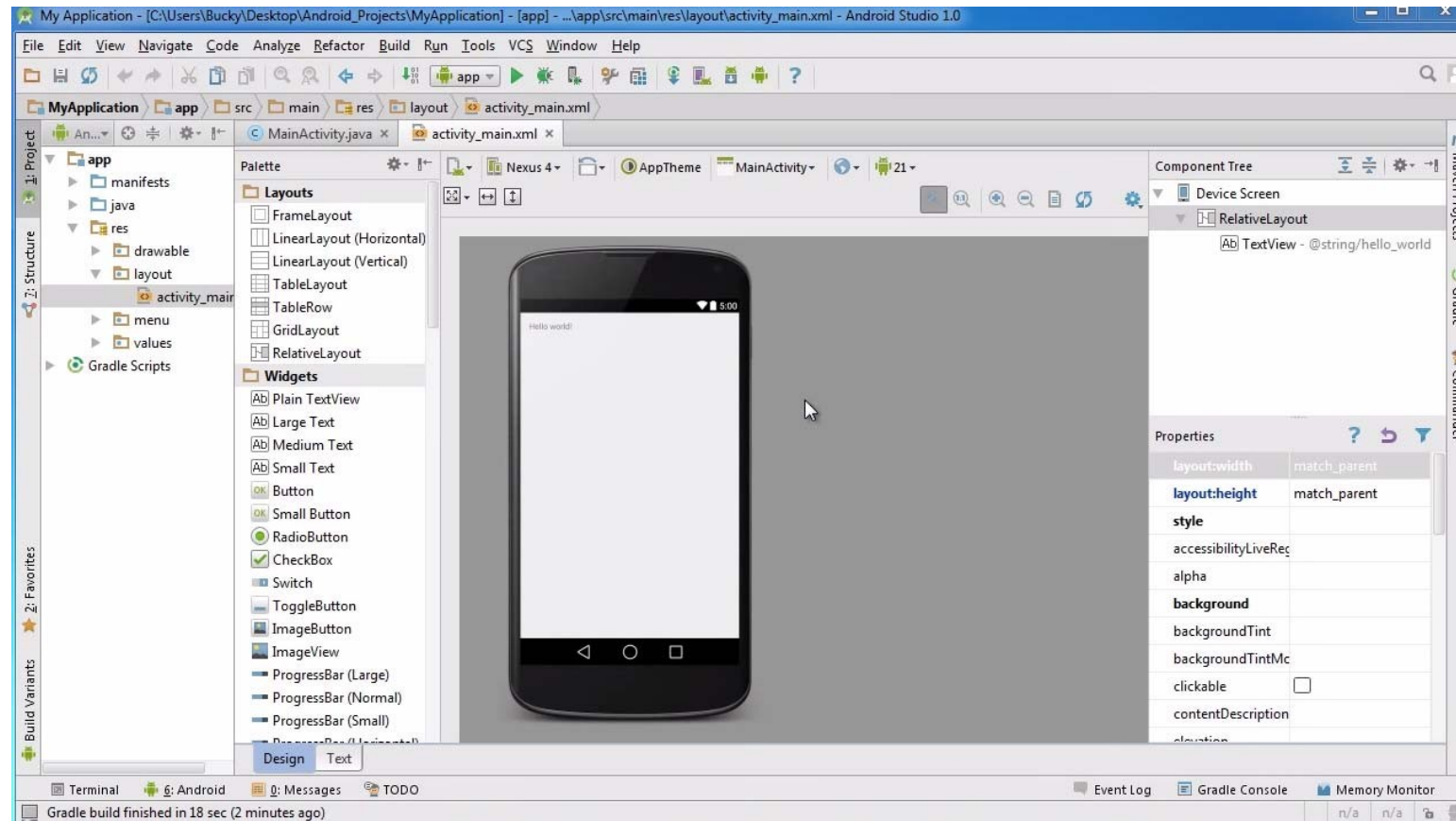
Setting up your Project

- After installing Android Studio, need to set up your project
- Tutorial: Android App Development for Beginners - 3
 - Setting up your project by Bucky Roberts (thenewboston)
 - <https://www.youtube.com/watch?v=r4olez0sfvY>
- Main steps to set up Android Project
 - Start a new Android Project
 - Configure new Android Project (select app name, domain name, etc)
 - Set platform and minimum SDK
 - Add an Activity

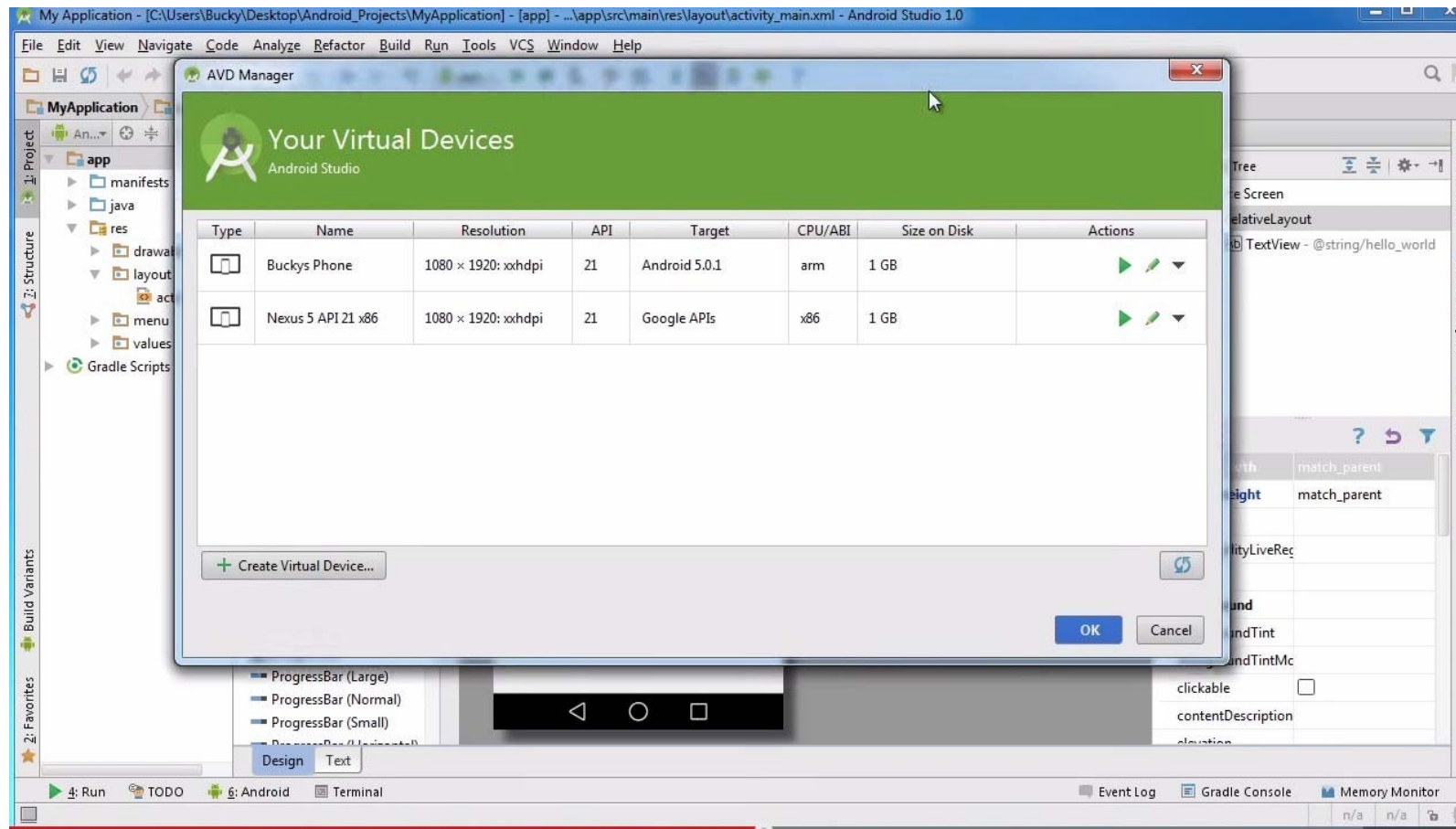
What is a gradle?

- When you click the run button in Android Studio, most of the actual work is done by an external build tool called Gradle.
- Gradle works behind the scenes to
 - Locates and downloads the correct versions of any third-party libraries you need.
 - Calls the correct build tools in the correct sequence to turn all of your source code and resources into a deployable app.
 - Installs and runs your app on an Android device.
 - A whole bunch of other stuff, like running tests and checking the quality of your code.

Android Studio

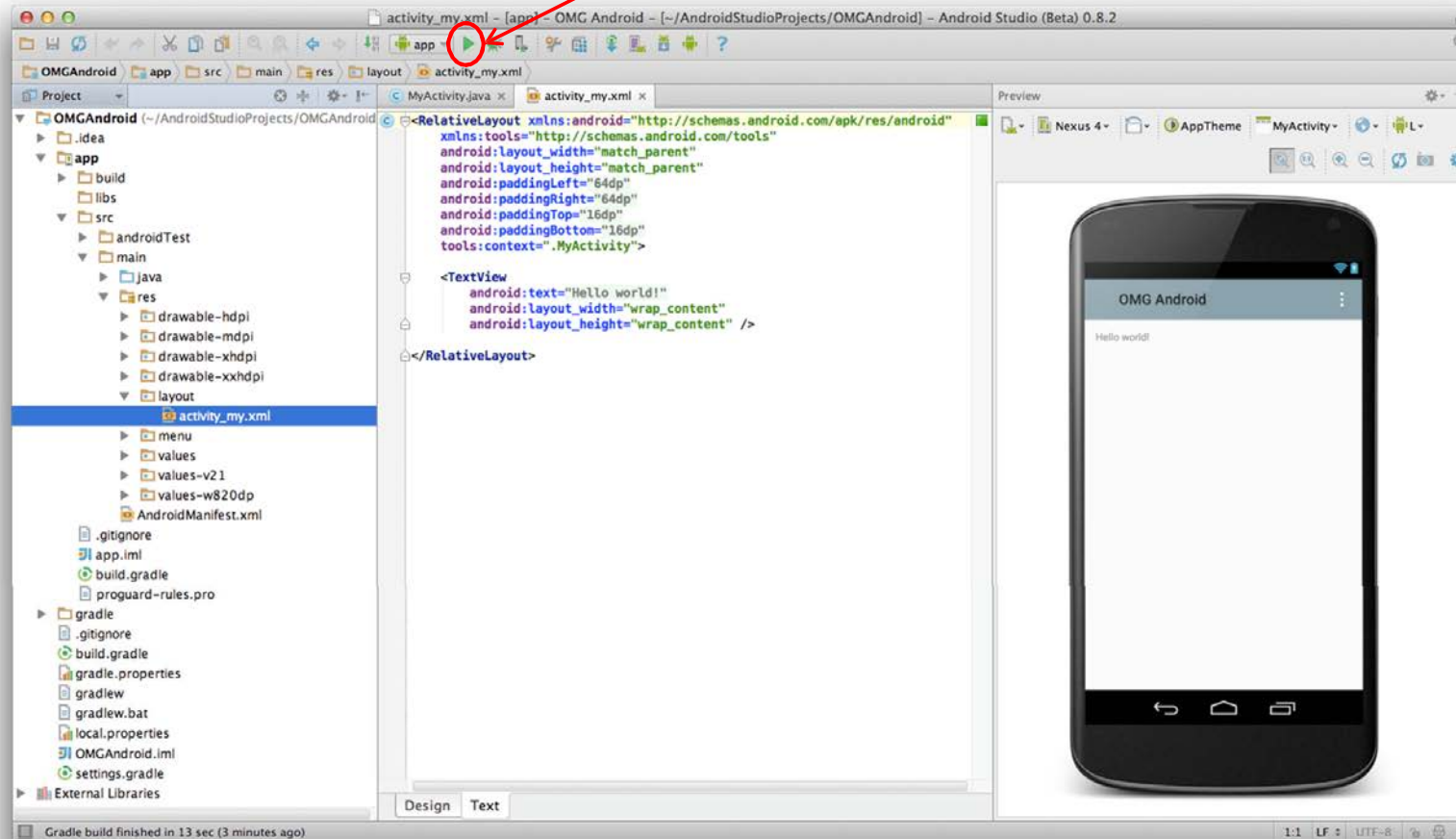


Run AVD Manager

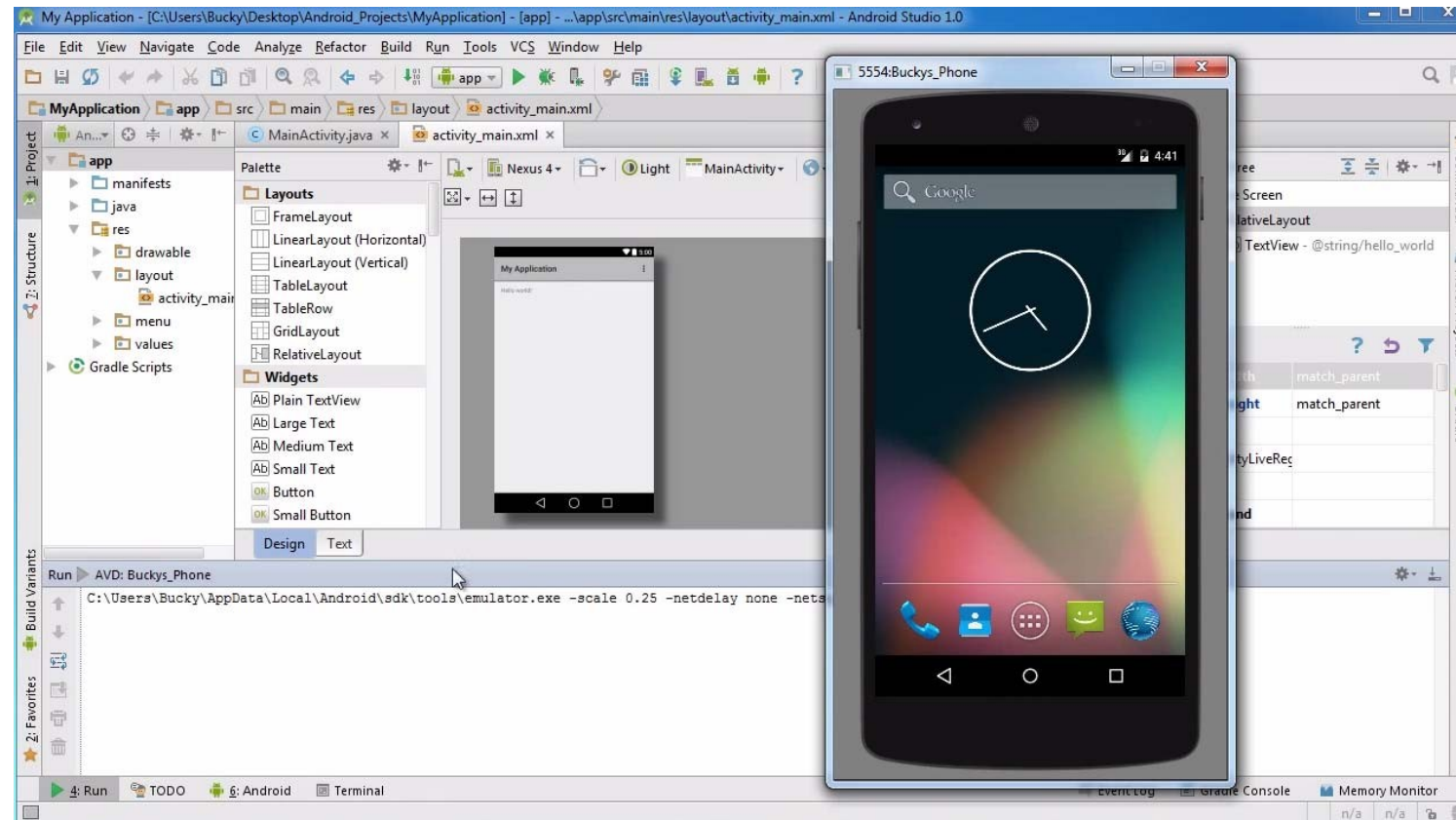


How to Run the App?

Click here to run the app



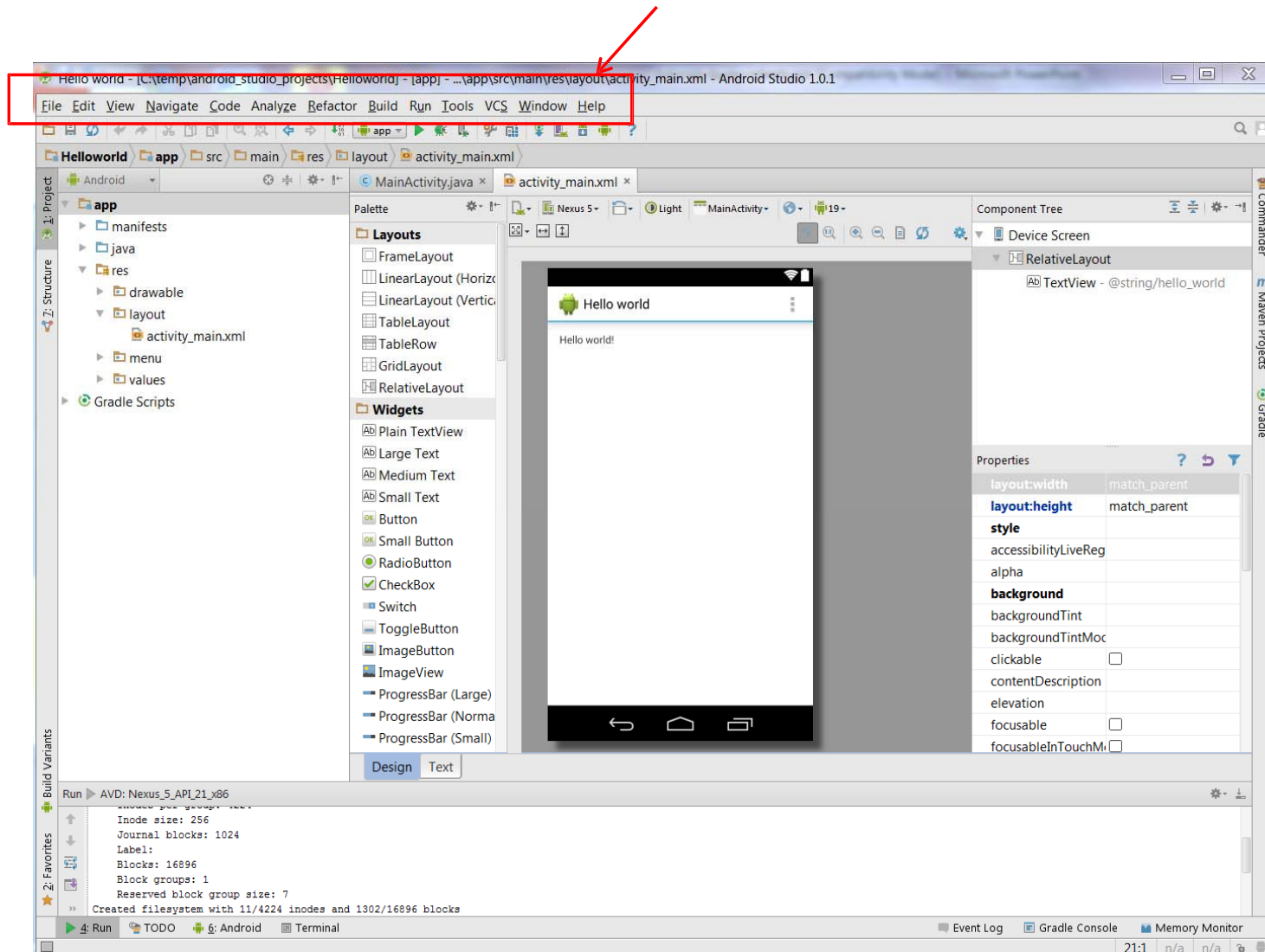
Run App on Virtual Device (Phone)



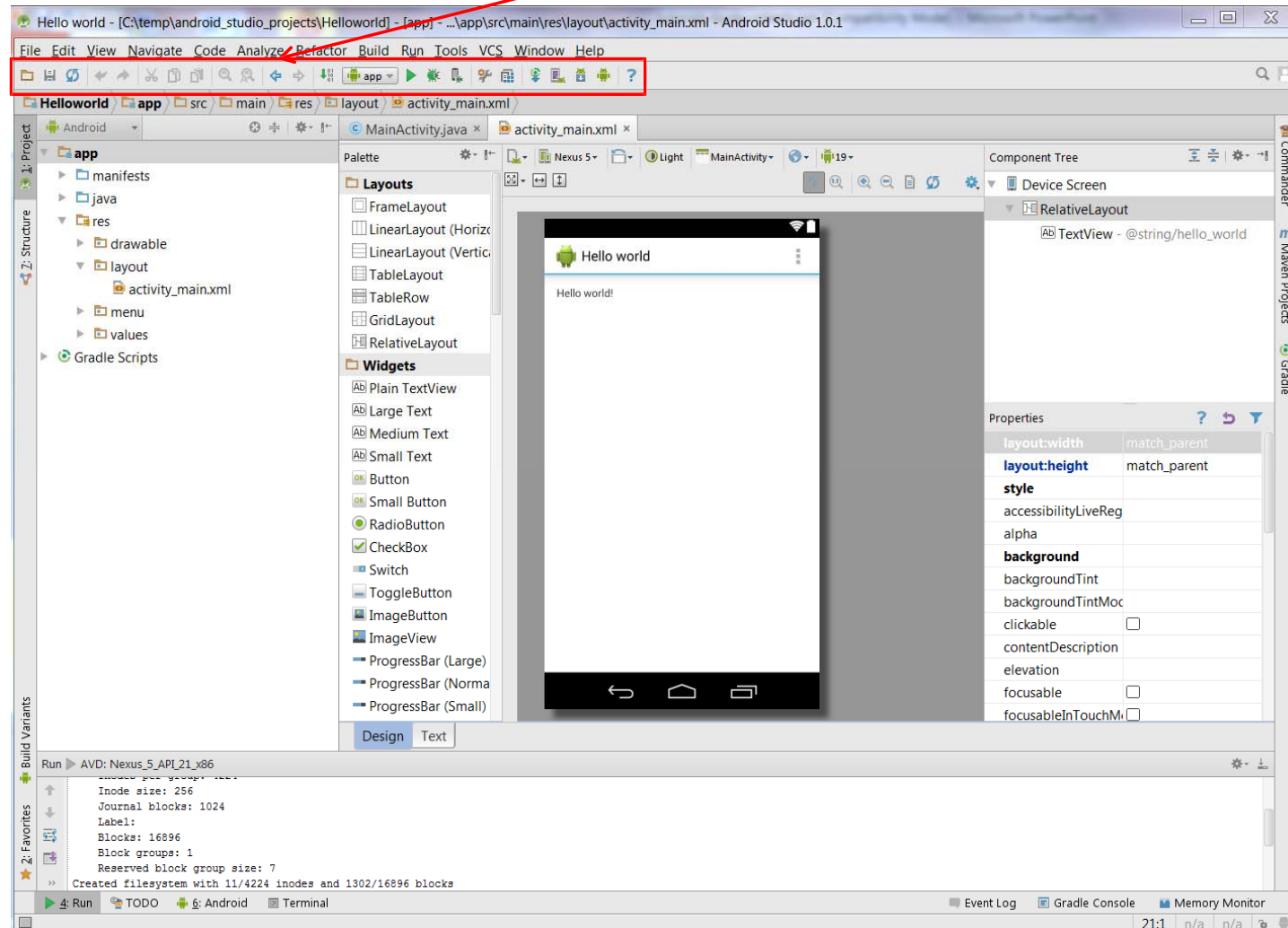
Tour of Android Studio Interface

- Tutorial 5: Tour of Android Studio Interface [6:01 mins]
 - <https://www.youtube.com/watch?v=-pdTqBq2TFQ>
- Quick overview of main sections of Android Studio
 - Windows menu bar
 - Android tool bar
 - Project window
 - Editor Window
 - Palette for Drag-and-Drop Design of Android buttons
- More detailed coverage of specific UI aspects later

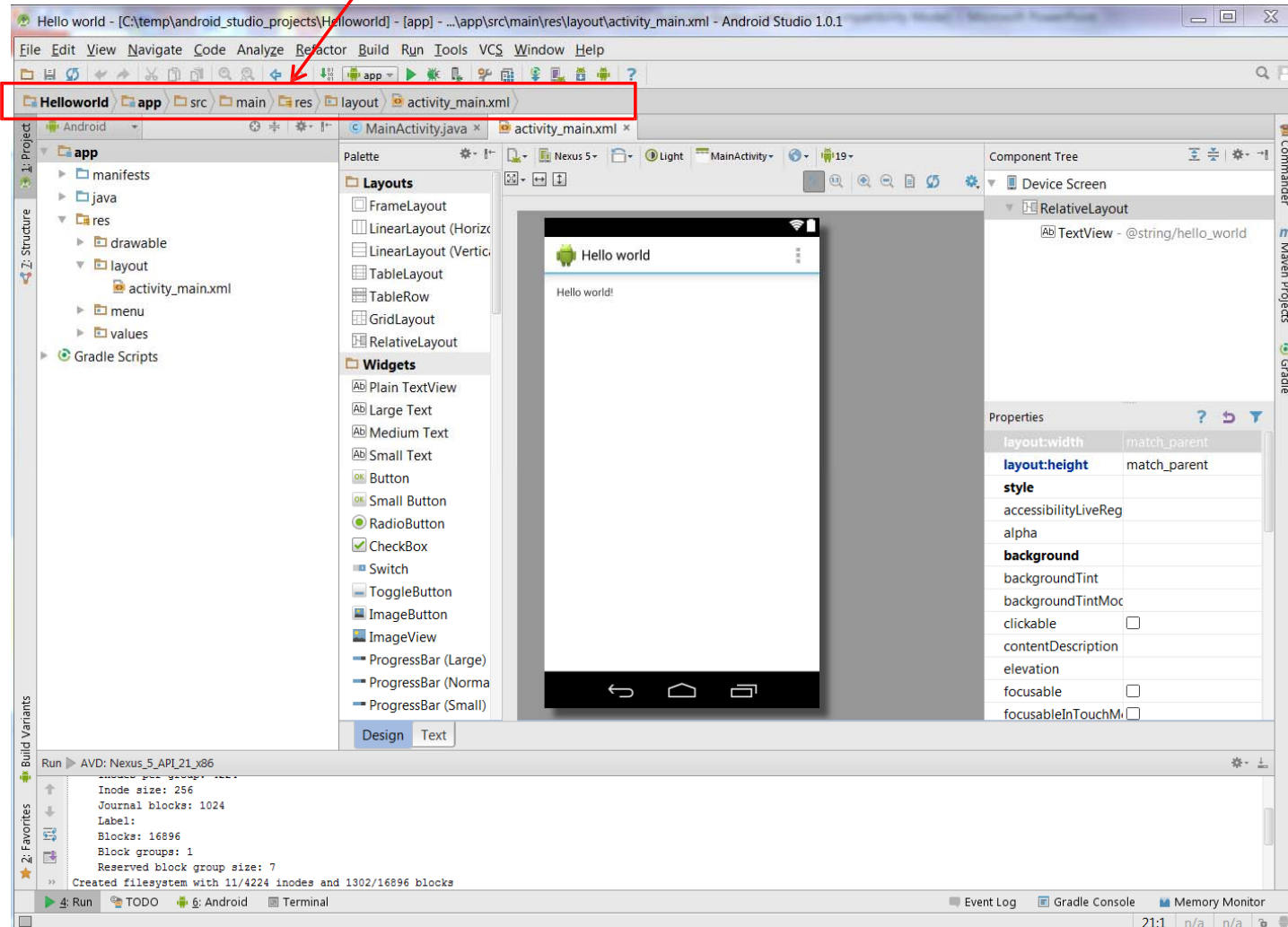
Typical Windows Menu Bar (File, edit, etc)



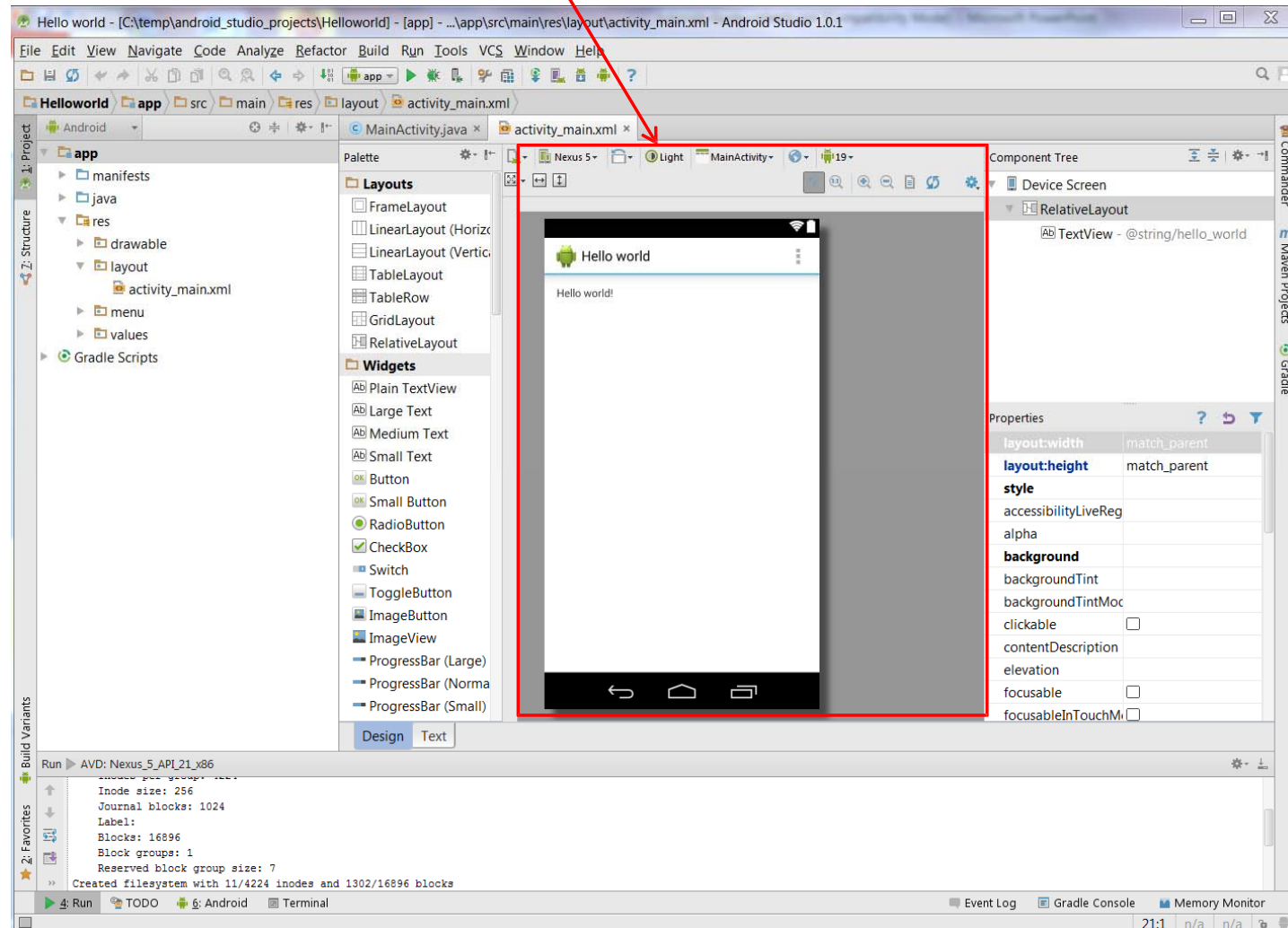
Tool Bar: Shortcuts to Frequently used Android-specific Functions (E.g. One-click access to SDK manager)



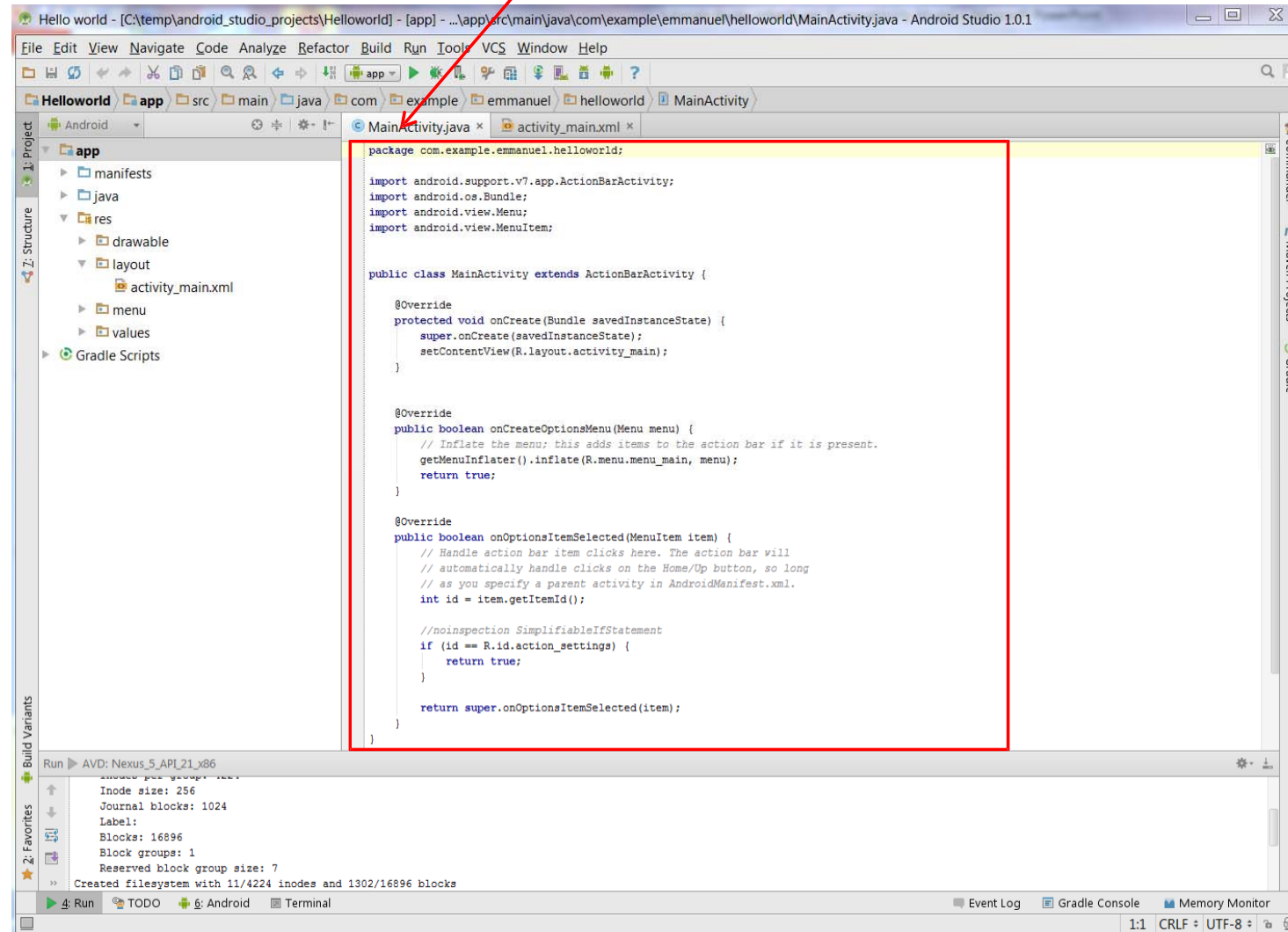
Path to Current File in IDE Window (Clickable)



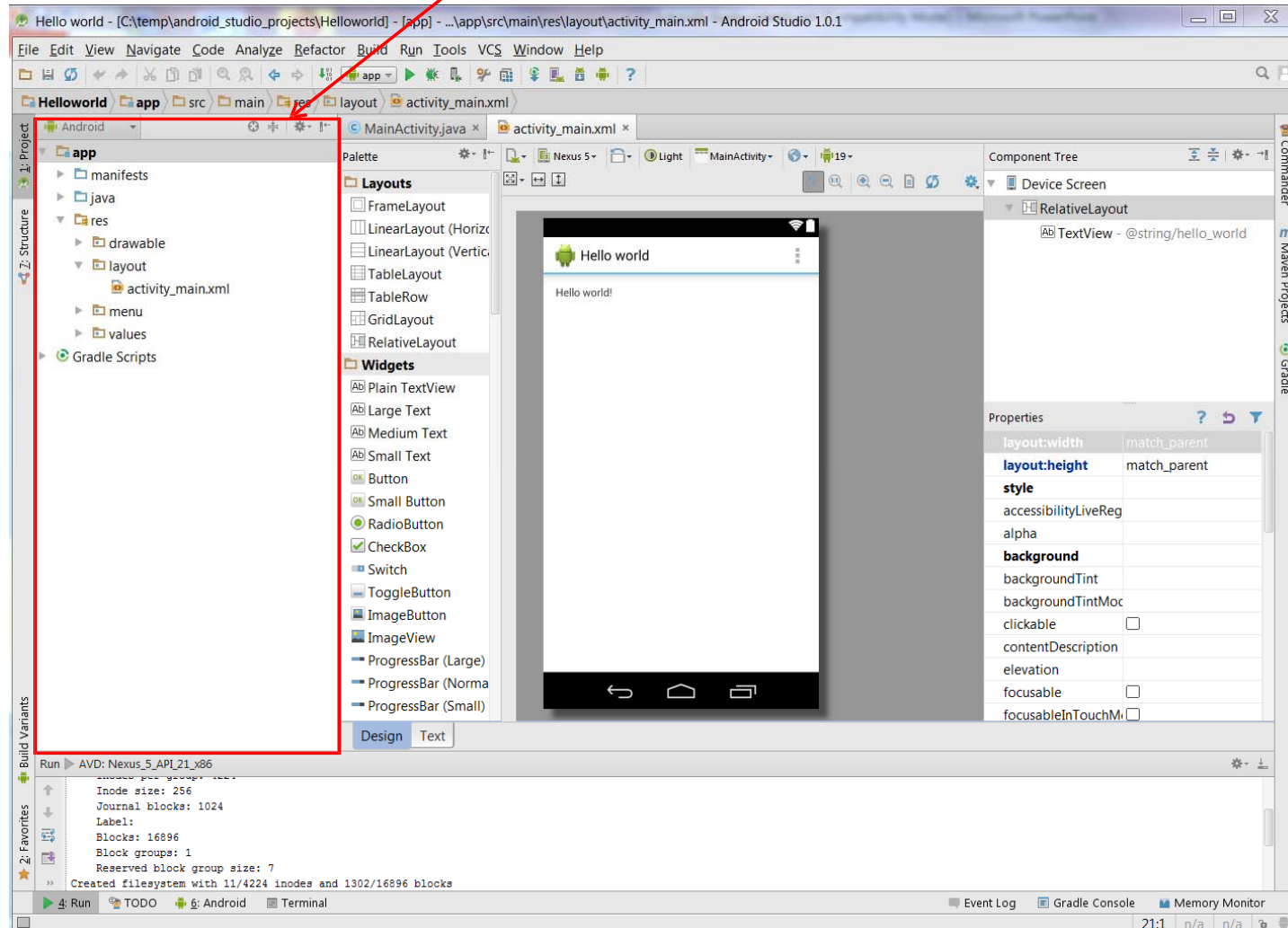
Editor Window (Allows editing of current file we are working on)



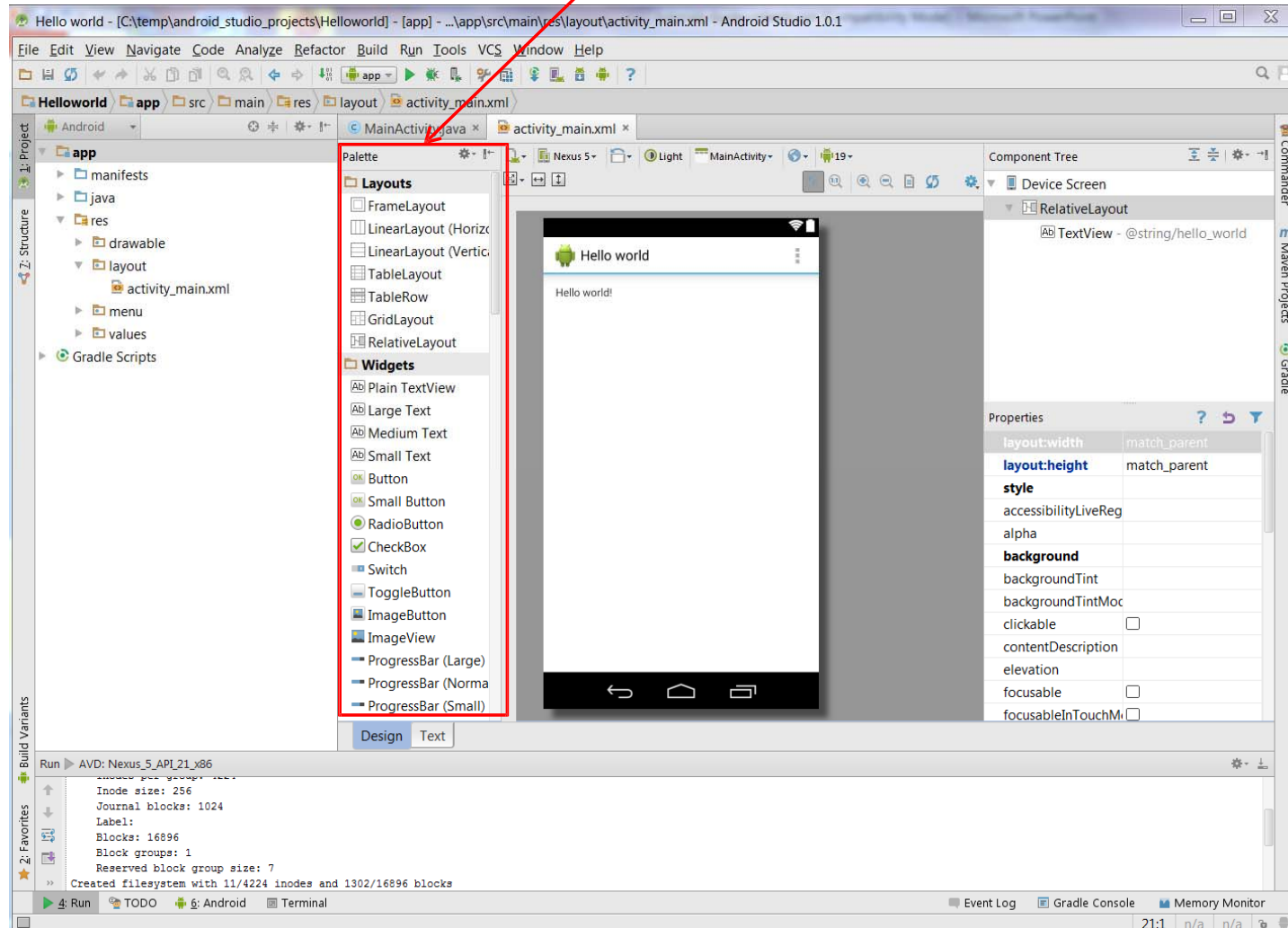
Clicking on Editor Window Tabs switches between Java code and Visual Interface



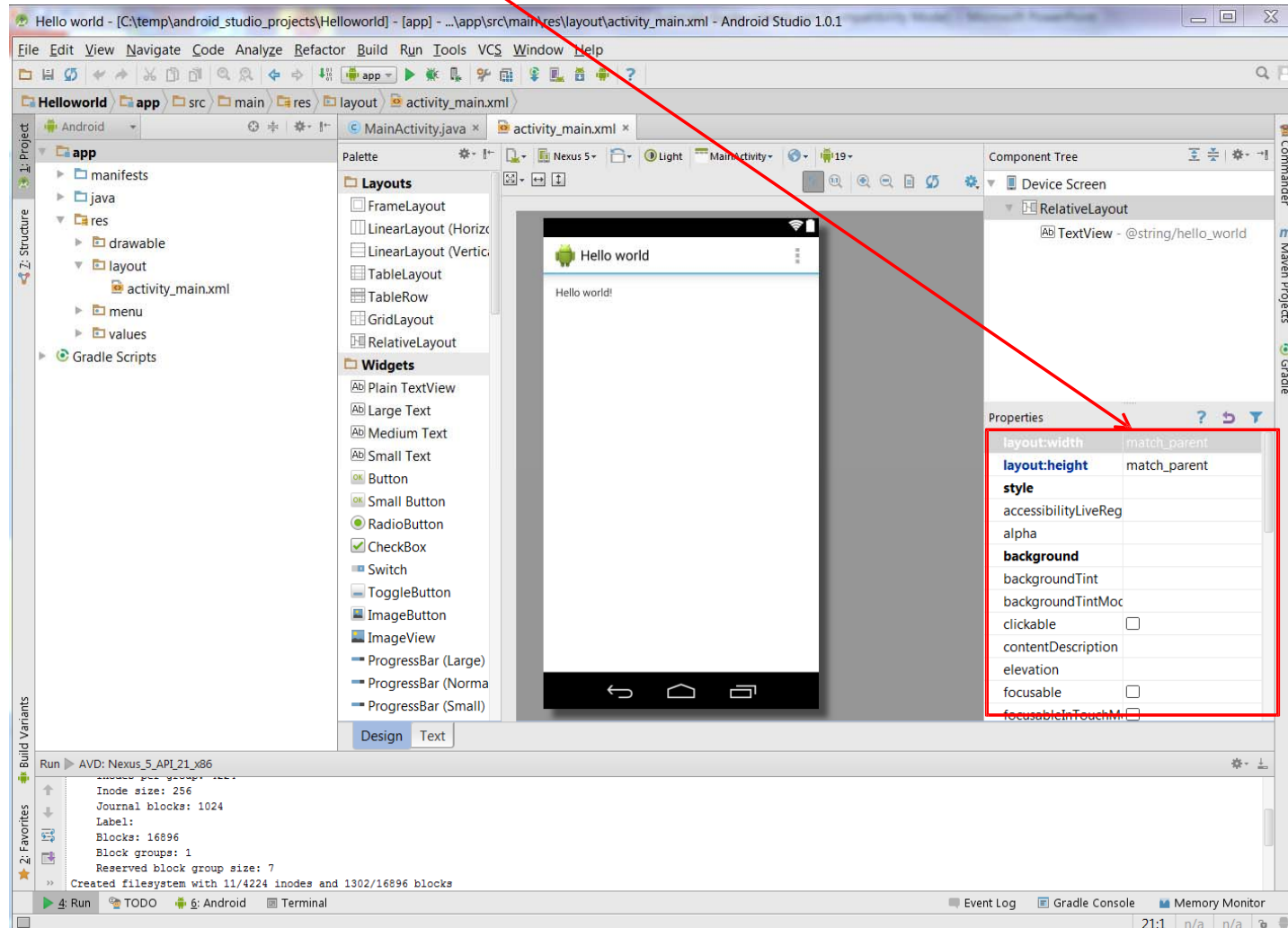
Project Window (Allows between project files, packages, etc)



Palette of Drag-and-Drop Elements for Designing Interface (Layout, widgets, etc)



Parameters of Drag-and-Drop Elements for Designing Interface (e.g. colors, dimensions of widgets, etc)



Importing Existing Code

- Can also import existing code
- The text comes with lots of free code you can learn from, use in projects as starting point
- Can import from gitHub repository
- See <https://developer.android.com/studio/intro/migrate>